

Technical Documentation

Diesel Engine
16 V 396 TE 74L

Description and
Operation Manual
M011135/00E

Valid from
year of Manufacture 1993



Printed in Germany

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Notes for the user

This Manual consists of 10 Sections, marked by tab-index sheets.

These Sections are listed on this page.

Each Section has its own detailed Table of Contents which lists the Groups and Subgroups.

The Groups in Sections B and G correspond to the Groups in the Spare Parts Catalog and other Product Support publications.

Using the Table of Contents to trace an entry

- ➡ Table of Contents (general)
- ➡ Table of Contents (section)
- ➡ Group / Subgroup
- ➡ Heading

Using the Alphabetical Index to trace an entry

- ➡ Term in Alphabetical Index
- ➡ Group / Subgroup
- ➡ Heading

Publication Amendment Service

Please complete and return the Confirmation of Receipt card; you will then receive amendments to your manual.

A Product Summary
Engine summary, technical data

B Description
System and subassembly descriptions

C Operating Instructions
Task descriptions

D Maintenance Schedule
Schedule and maintenance frequency chart

E Troubleshooting
Fault conditions, probable causes, remedies

F Drawings
Engine cross-sections

G Task Description
Inspection and maintenance

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A.010	Engine summary Engine model designation Engine power Engine layout - cutaway drawings
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Diesel Engine

Part A
Product Summary

The product summary gives details of the engine and contains technical data such as engine power, general engine specifications, engine weight, oil and coolant capacities, settings and operational data.

ENGINE LAYOUT, ENGINE POWERKey to Engine Model Designation 16 V 396 TE 74L

16	=	number of cylinders
V	=	Vee configuration
396	=	engine series - 100 times displacement of one cylinder in litres
T	=	exhaust turbocharging
E	=	intercooling in split-circuit coolant system
7	=	marine engine
4	=	design index
L	=	uprated

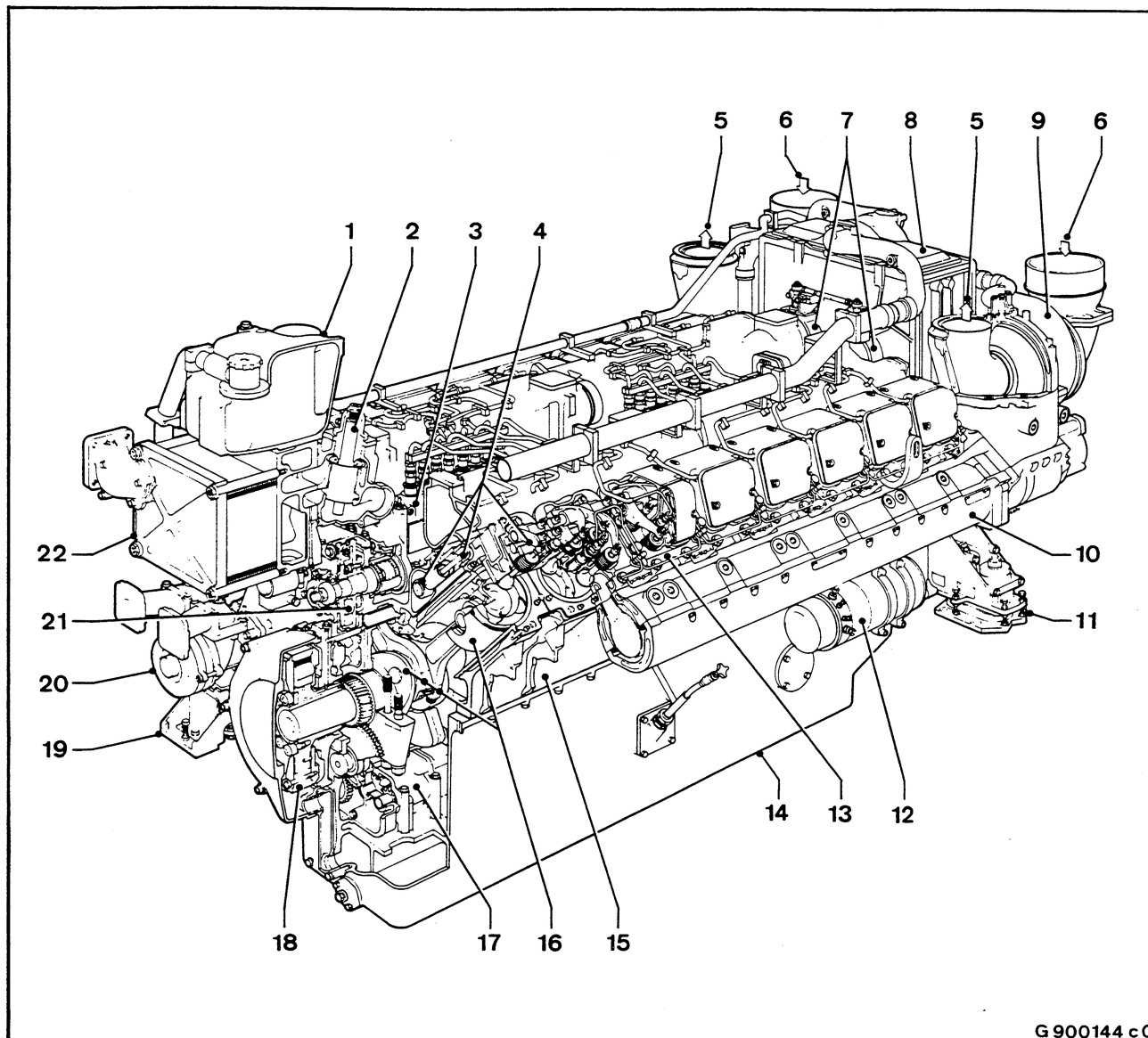
Engine Power

Continuous fuel stop power 2000 kW at 1900 rpm

Reference conditions

- intake air temperature	25 °C
- raw water temperature	25 °C
- barometric pressure	1000 mbar

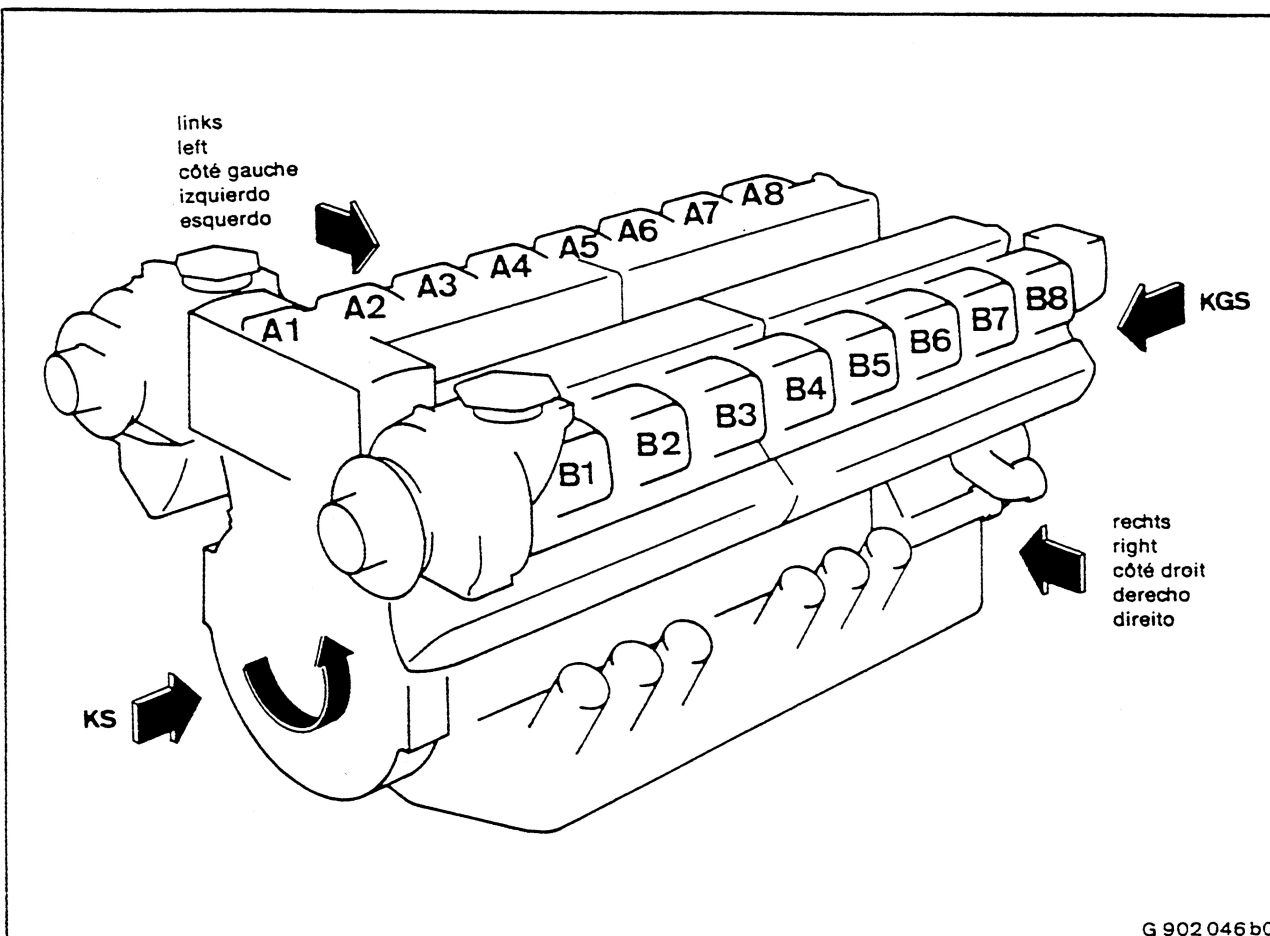
Engine Layout Cutaway Drawing 16V 396 TE.4 (also applies to 8 and 12V)



G 900144 c 0

- | | |
|-------------------------------|-----------------------------|
| 1 Coolant expansion tank | 12 Starter |
| 2 Coolant thermostat | 13 Cylinder head |
| 3 Fuel injection pump | 14 Oil pan |
| 4 Valve gear | 15 Crankcase |
| 5 Exhaust outlet | 16 Running gear |
| 6 Air inlet | 17 Engine oil pump |
| 7 Emergency air shut-off flap | 18 Vibration damper |
| 8 Intercooler | 19 Engine mount, timing end |
| 9 Exhaust turbocharger | 20 Bilge pump |
| 10 Exhaust manifold | 21 Fuel injection timer |
| 11 Engine mount, flywheel end | 22 Coolant cooler |

ENGINE SIDE AND CYLINDER DESIGNATIONS



Engine sides are always designated as viewed from the main PTO end.

The left bank cylinders are designated "A" and the right bank "B" (according to DIN Standard No. ISO 1204).

The cylinders of each bank are numbered consecutively, starting with No. 1 at the main PTO end.

Numbering of other engine components is also from the main PTO end, starting with No. 1.

Designations and Abbreviations

Main PTO end = KS
 Auxiliary PTO end = KGS
 Left side
 Right side
 Top
 Bottom

GENERAL SPECIFICATIONS

Operating method	four-stroke cycle, single-acting
Combustion method	direct injection
Mode of supercharging	exhaust turbocharging single-stage sequential turbocharging
Cooling method	liquid-cooled
Cylinder arrangement	90°Vee
Bore	165 mm
Piston stroke	185 mm
Displacement, cylinder	3.96 litres
Number of cylinders	16
Displacement, total	63.3 l
Compression ratio	12.3 : 1
Direction-of-rotation (viewed from driving end)	c.c.w. (non reversible)
Firing order	A1 - B5 - A3 - A5 - B2 - B8 - A2 - A8 - B3 - A7 - B4 - B6 - A4 - A6 - B1 - B7
Nozzle opening pressure:	
- setting pressure with new spring	280 bar to 288 bar
- specified value with run-in spring	270 to 278 bar
- minimum pressure with run-in spring	260 bar
Final compression pressure at 120 rpm, 1000 mbar and engine coolant temperature of 5 °C	20 to 24 bar
Breakaway torque at 40 °C engine coolant temperature, allowance for acceleration included	approx. 800 Nm
Cranking torque at engine coolant temperature of 40 °C	approx. 750 Nm
Firing speed at engine coolant temperature of 40 °C	approx. 120 rpm
Mean piston speed	11.7 m/s at 1900 rpm
Engine noise mean free-field level	109 dB(A) 1 m distance
Exhaust noise, undamped mean free-field level	118 dB(A) 1 m from outlet